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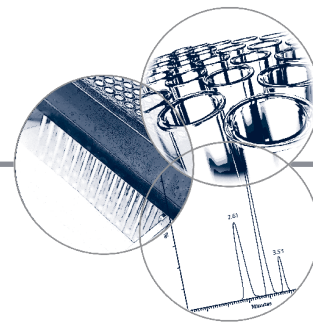
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Perspectives on addressing ionization matrix effects in LC–MS bioanalysis

Ionization matrix effects are one of the most difficult issues in LC–MS bioanalysis that are without a good and universal solution. Most people in the field are aware of it, but some are not sure how to deal with it. Many laboratories do not routinely assess matrix effects in method validation or substitute it with a different experiment that only indirectly and partially test matrix effects. Others, when matrix effects are mentioned, immediately link them with phospholipids and try to use means to remove them from the samples (e.g., SPE), without understanding if the phospholipids can really impact the analytes. In this article, key issues related to matrix effects will be examined, and methods to address matrix effect problems will be evaluated for effectiveness and practicality.

Over the past 20 years, LC–MS/MS has transitioned from being an ‘elite’ technique to a routine tool in quantitative bioanalysis. There has been tremendous progress in instrument performance and ease of use that facilitated such a transition. Nevertheless, one major drawback in this technique, namely **matrix effects** (ME) on ionization, remains the Achilles’ heel [1–4]. The progress in the understanding of ME and its impact on data quality in LC–MS bioanalysis over the last 12 years is evident by looking at the content of the 2001 US FDA guidance for Bioanalytical Method Validation and the conference report of the 3rd AAPS/FDA Bioanalytical Workshop held in 2006 [5–6]. In the 2001 FDA guidance, discussion of ME on LC–MS analysis was briefly mentioned without giving a detailed instruction on how to assess ionization matrix effects. In the 3rd AAPS/FDA Bioanalytical Workshop conference report, a section was devoted to the assessment of ME. Given the increased understanding of ME in LC–MS bioanalysis, the approach to ME assessment in method development/validation remains quite diverse, to say the least. Many laboratories still do not conduct a full evaluation of ME during method validation, relying on the selectivity test by spiking six blank matrix lots at the LLOQ as a substitute for ME assessment. While such practice provides some assurance for the method not to be susceptible to matrix influence in concentration measurement, it does not give any information on whether ionization ME occur in the method.

Part of the reason why many people do not perform a quantitative assessment of ME is

that there is no such stipulation in the current FDA guidance. Although the 3rd AAPS/FDA Bioanalytical Workshop conference report contains a section on quantitative assessment of ionization ME, people are either not familiar with it or do not consider it a requirement. Another reason given for some laboratories not conducting the quantitative measurement of ME is that there are difficulties in carrying out the experiment in performing such a measurement, which will be discussed later. The common method of quantifying ME is based on the comparison of the analyte signal (e.g., peak areas) of a neat solution versus that of the analyte spiked into blank matrix extract. Using this experiment, Buhrman *et al.* devised the term ion suppression ($\text{ion suppression} = 100 \times [\text{neat} - \text{spiked}] / \text{neat}$) to quantify ME [3]. Matuszewski *et al.* suggested the term ‘matrix effect’ ($\text{ME} [\%] = 100 \times \text{spiked} / \text{neat}$) [7]. Bansal and Liang coined the term **matrix factor** ($\text{MF} = \text{spiked} / \text{neat}$) [8]. The Crystal City conference report contains a clear explanation on the use of MF for method validation. All three methods provide similar information about the quantitative aspect of ME while MF is a succinct and unequivocal term in describing the quantitative aspect of ME.

Another area of ME in LC–MS bioanalysis with divergent views is how much attention should be given to examining ME, at what stage of the project. It is obvious that more emphasis on evaluating ME has been given in regulated bioanalysis, primarily due to the requirement of the FDA guidance and the Crystal City conference report. Bioanalysis in

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Key Terms

Matrix effects: Suppression or enhancement of ionization of analytes in LC–MS analysis.

Matrix factor: A term used to quantitatively represent the extent of matrix effects, defined as the response of the analyte in the presence of matrix divided by that in neat solution.

Internal standard: An analyte spiked to samples at a fixed concentration in the analytical procedure to correct and monitor perturbations during analysis.

a drug-discovery setting does not usually look into the impact of ME until a large discrepancy from expected pharmacokinetic (PK) results are observed. In reality, discovery bioanalysis may encounter more severe matrix effects owing to the generic nature of the methods and the lack of stable isotope-labeled **internal standards** (IS), especially for intravenous (iv.) PK studies. For many who are concerned about ME, much emphasis has been placed on phospholipids [9–14]. The efforts in solving potential ME interference from phospholipids are helpful for the cases when analytes co-elute with phospholipids. Nevertheless, ME were more frequently observed due to other known or unknown components. Addressing ME from nonphospholipids may be of a more common need than from phospholipids.

Source of ionization ME

■ Phospholipids & other unidentified components from biological matrices

When ME are mentioned in bioanalysis, phospholipids are most frequently associated with them in many people's minds. In the past 15 years performing pharmaceutical bioanalysis, we have seen only a handful of cases where phospholipids cause ME issues. The compound diversity in our analysis covers a large spectrum in terms of therapeutic areas (oncology, virology, autoimmune and metabolic diseases), with $\log P$ values covering the range -2 to 9. One reason for such 'good luck' is that the compounds in

pharmaceutical R&D are most often separated from the phospholipids in a typical reversed-phase LC–MS method using a C_{18} column. Most pharmaceutical compounds in our diverse therapeutic areas tend to elute earlier than the first phospholipid in the matrix. This is illustrated in **Figure 1**, which is the MRM chromatogram of a rat plasma extract containing 100 ng/ml saquinavir. The HPLC conditions used were: mobile phase A was water with 0.1% acetic acid; mobile phase B was acetonitrile with 0.1% acetic acid; an Ascentis Express C_{18} column (2.7 μm ; 20 $\times$ 2.1 mm) running a gradient at 0.5 ml/min from 30 to 90% B in 0.8 min, then to 100% B from 0.8 to 1.2 min, held at 100% B until 1.9 min, returned to 30% B at 2 min and equilibrated. Saquinavir is a medium retention compound among typical pharmaceutical R&D compounds on a C_{18} column, eluted at 0.57 min under the above conditions at about 70% organic composition. The earliest eluting phospholipids (lysophosphatidylcholines) eluted after 1.1 min, at above 90% organic. These phospholipids were called 'mid-eluters' by Bennett and Van Horne, probably due to the reduced retention under the HPLC conditions used with a phenyl-hexyl column [9]. However, on C_{18} columns, the retention times of these mid-eluters are later than the large majority of pharmaceutical compounds currently worked on. This could be one major reason why we infrequently saw problems in discovery and development support due to phospholipids. Our observation is supported by the fact the desirable physicochemical properties for pharmaceutical compounds are such that they usually fall into mid-eluters region on a small C_{18} column used in LC–MS bioanalysis (eluting between 20 and 70% acetonitrile content under shallow HPLC gradients). For a period of time, we put in a MRM trace to monitor phospholipids in our discovery bioanalysis. We stopped this practice, however, after we found that most of our compounds eluted earlier than phospholipids. In light of our observation, the attention given to phospholipids in ionization suppression in pharmaceutical bioanalysis is excessive. To further illustrate this point, **Figure 2** shows a total ion chromatogram (TIC) and a mass spectrum of the same rat plasma sample obtained by full scan. In the chromatogram (top trace), one can clearly see a big rise of total ion signals after a retention time of 1 min, resultant from the elution of matrix components that can cause ionization suppression. When one sums up the mass spectra from 1.1 to 2 min, it results

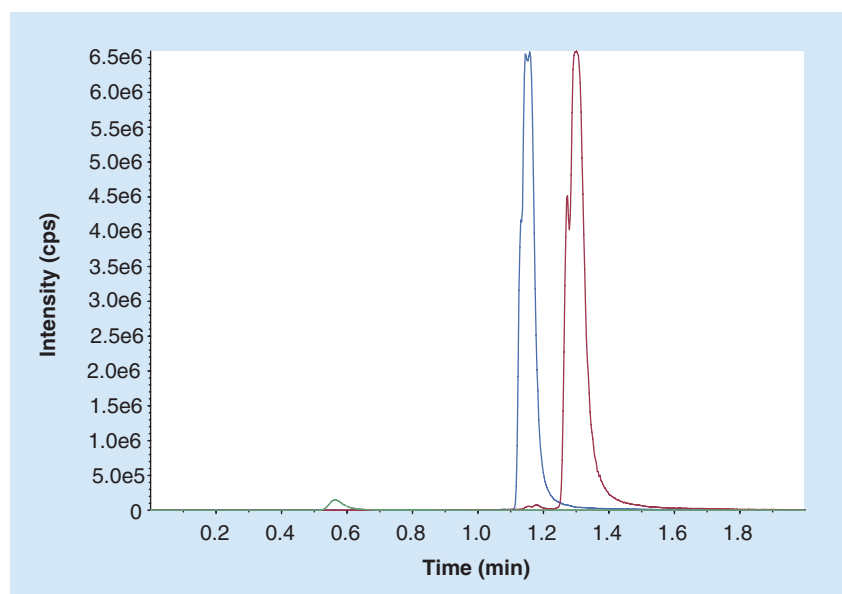


Figure 1. Rat plasma extract containing saquinavir. The peak at 0.57 min is from saquinavir, where the peaks at 1.17 and 1.31 min are from the two earliest eluting phospholipids.

in a mass spectrum in the bottom trace. In this mass spectrum, in addition to the half dozen known phospholipids (e.g., m/z 496, 524, 758 and 786), one can see tens or more ions across the whole mass range of the spectrum that can cause suppression for analytes eluting in this late-retention region. It should be noted that in the high-organic elution regions, one may not observe intense ion peaks yet still experience significant suppression from the co-eluting, not easily ionizable components. This is what was frequently observed in methods for late-eluting analytes. The point is that phospholipids are only a few of the matrix components in the nonpolar region that may cause ME problems in a bioanalytical method. For this reason, when an isocratic LC–MS method is performed, it is critical to employ a column clean-up step at the end of elution by ramping up the mobile phase to high organic, to avoid bleeding of accumulated nonpolar components that come out in subsequent injections, which cause ME.

Even though phospholipids bore more than justified blame for causing ME problems in LC–MS bioanalysis, they are sometimes indeed the problems for many late-eluting analytes. Of two papers describing the removal of phospholipids by SPE, the one using Sigma-Aldrich's zirconia-based HybridSPE provided better selectivity for preferential depletion of phospholipids [11,12]. When using conventional SPE to remove interfering phospholipids from analytes, the challenge is to select the elution conditions that differentiate the analytes with similar polarity as the interfering phospholipid. Resolving phospholipid suppression by HPLC separation seems to be, if not more efficient, then definitely more economical in comparison with regular SPE [13]. In our laboratory, we normally adjust HPLC conditions to solve the problem of ME before we attempt different extraction conditions. Frequently, it is unnecessary or economically unwise to use SPE to remove phospholipids in pharmaceutical bioanalysis, for reasons discussed above.

■ Intravenous formulation excipients

Polymeric iv. dosing excipients cause many more ME problems in pharmaceutical R&D. We showed that surfactants such as polyethylene glycols and Cremophor® in iv. formulations cause ME for iv. PK samples [15]. Interference from these excipients is more difficult to address as polymeric components elute across a wide retention window, hovering in the intermediate

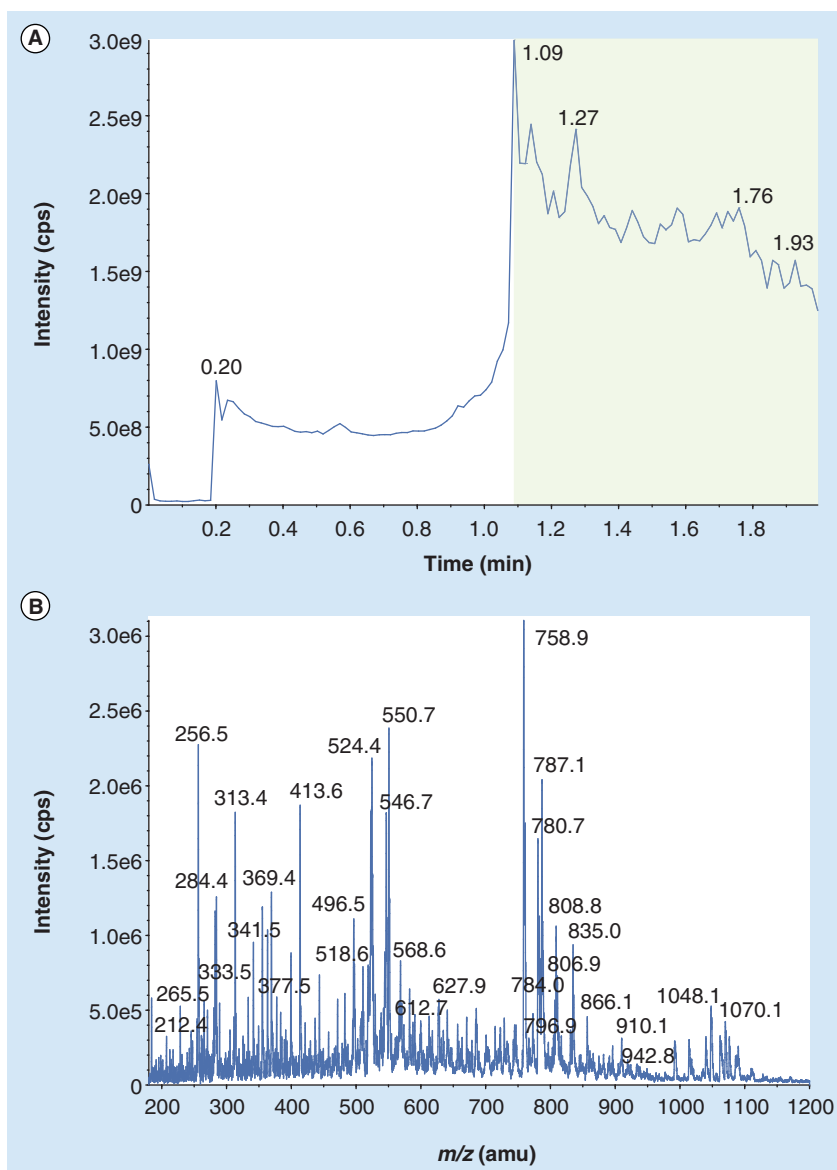


Figure 2. Ion chromatogram and summed mass spectrum over a retention region from a rat plasma sample. (A) Total ion chromatogram. **(B)** Summed mass spectrum of retention region 1.1–2 min.

retention time range. Because of this difficulty, some people try to avoid using these surfactants in iv. formulations. The dilemma, especially for drug discovery, is either you spend time in developing an iv. formulation without surfactants or you put in the effort to avoid or solve ME problems in bioanalysis. In our laboratories, we decided not to limit what the galenic group can use in formulation and, therefore, have to contend with potential ME issues in discovery bioanalysis. Typically, ME from the surfactants occur in samples of early timepoints, for example, up to 1 h after dosing. Later timepoints typically contain these surfactants at lower concentrations

that cause little matrix effects. To prevent ME in early timepoint samples, one can dilute these samples by tenfold or more with control plasma, which reduces ME from the surfactants. Even after diluting the early timepoints, one cannot be certain that ME have been removed. Huang suggested a way to monitor ME for the early timepoint samples, using another aliquot of the early timepoint samples analyzed at a higher dilution [HUANG R. PERS. COMM.]. If the two measurements with different dilution factors agree with each other, it indicates ME is insignificant. If the two measurements for the early timepoints differ by more than a certain threshold (e.g., 30%), one may need to improve the method and reanalyze samples. Recently, we used this approach to monitor a couple dozen iv. PK studies and did not find a ME issue related to iv. samples containing PEG. Of course, one can always resort to selective extraction methods, including liquid–liquid extraction, SPE and HPLC to tackle ME problems from surfactants. The downside of such approaches is the limited throughput.

Methods of ME assessment

As mentioned in the introduction, assessing ME is the least standardized step for bioanalytical method validation. It is understandable that assessing ME routinely in drug discovery may not be justified from a cost–benefit standpoint; however, it would be imprudent to say the least, not to know about ME in a method for regulated bioanalysis, especially when the method does not use stable-isotope IS. Although ME assessment is not considered an essential parameter in method validation, one can easily argue that the determination of MF or its equivalent is more important than the determination of extraction recovery, which is a required parameter during validation [5]. Nevertheless, measuring MF can be a tricky experiment. The conventional way of comparing the response of the analyte in a neat solution versus that of the analyte spiked into control matrix extract sometimes presents a practical problem. First, many compounds exhibit nonspecific binding in neat solutions at low concentrations. We have seen many cases where measuring MF by the spiking method was not feasible because analytes were completely lost during the experiment, either in the sample vial or in the HPLC flow path. Such nonspecific binding is less common for analytes spiked in control matrix extract, because the matrix components coat the active surface areas that may

cause nonspecific binding. Moreover, measuring MF by the spiking method sometimes results in distorted values due to nonspecific binding. We have found that some ionization enhancement results were due to nonspecific binding rather than actual enhancement. The worst part is how one can ascertain that the determination by the spiking method is free of nonspecific binding distortion. Naturally, one may find ways to minimize or eliminate nonspecific binding, for example by modifying solvent composition, adding modifiers to HPLC mobile phase and solutions, or by coating sample containers with blocking agents. Nevertheless, finding a solution to nonspecific binding problems in the spiking method is restricted by the fact that the components added to solve the problem may themselves cause ME. There is another drawback with respect to the spiking method; one cannot obtain a true control extract in which to spike the analyte for an online extraction method, and therefore the method is not amenable to online extraction methods.

We have developed a method, termed post-column infusion (PCI) profile convolution to measure MF, which is free of the two problems described above [16]. This method utilizes the PCI experiment developed by King *et al.* [4] and applied a convolution step to extract quantitative information out of the PCI profile. Traditionally, people use a PCI profile by injecting a blank matrix sample alone to qualitatively assess ME. The perturbation of the PCI profile provides some sense of ionization enhancement or suppression, especially by comparing the PCI profile of blank matrix against that of a solvent blank. But looking at the PCI profiles alone does not provide quantitative measurement of ME. When a PCI profile is weighted by a chromatogram of the analyte (a point-by-point multiplication of an infusion profile with the normal analyte chromatogram, which can be viewed as a digital form of convolution), the resulting trace resembles a typical chromatogram, illustrated in **FIGURE 3**. The peak area in the convoluted profile (product trace) correlates to the response of the analyte in the presence of matrix of the sample injected to obtain the infusion profile. Therefore, by comparing the areas of the convoluted PCI profile from injecting a control matrix blank versus that from a solvent blank, one can measure the MF of the analyte. Because PCI profiles of any methods, including online extraction, can be obtained, this PCI profile convolution works for any methods. In addition, the nonspecific

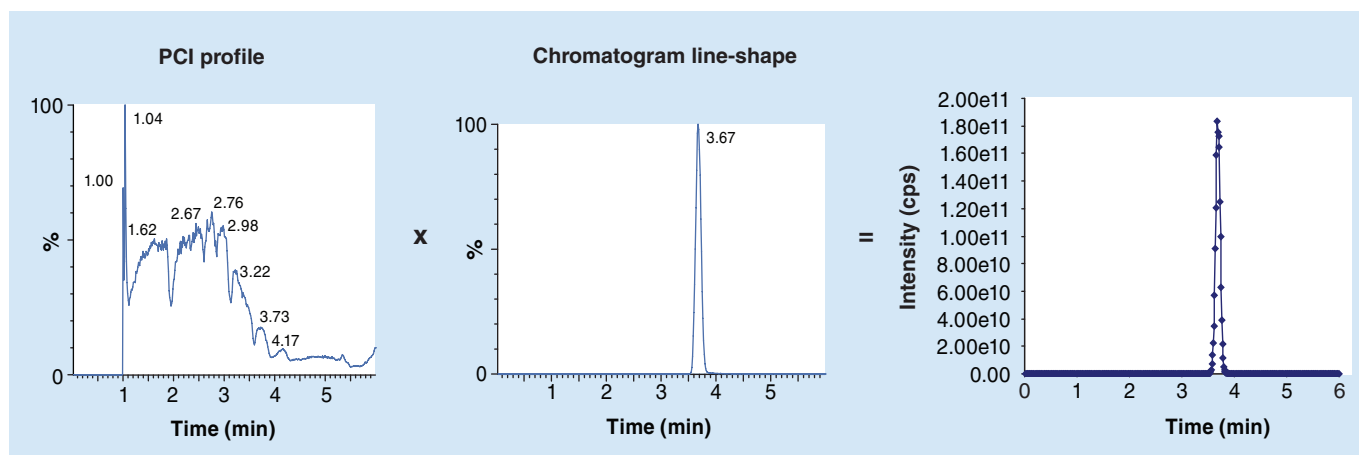


Figure 3. Post-column infusion profile convolution by sequentially multiplying the intensity of the two traces at each retention point.

binding problem sometimes seen in the spiking method mentioned above is avoided because the analyte in the infusion syringe is at a much higher concentration. In the worst case scenario, when nonspecific binding is seen in the infusion syringe, a remedy is easy to obtain, by either using a coated syringe or having the syringe saturated with analyte before taking up the infusing solution. One thing worth pointing out in obtaining a PCI profile is that for a method with high LC flow into the mass spectrometer, the high back-pressure to the infusion syringe results in delay for the infusion to reach a steady flow, when the flow is switched from divert-to-waste to the mass spectrometer (commonly employed in bioanalytical methods), resulting in skewed profiles at retention times close to the valve-switching event. Removing the divert step to obtain PCI profiles will avoid this problem, but introduces artificial ME for the early retention region, due to the void volume components being flowed into the ion source. One way to solve this high back-pressure problem is to either use a larger capillary on the source probe to reduce back-pressure or to use a make-up flow at the divert valve to maintain a constant pressure. Once PCI profiles are obtained, the remaining challenge is to perform the digital convolution. Fortunately, two major MS vendors, AB Sciex and Waters, have implemented into their software (Analyst and MassLynx™) the procedure to perform calculations related to PCI profile convolution. This PCI profile convolution method facilitates a means to reliably measure MF under any foreseeable situations.

One other question that remains is whether we need to measure MF at different analyte

concentrations. Typically, if the matrix is constant, the ME from that matrix is uniform across the concentration range of the calibration curve, evident by the linear calibration curves routinely seen even when ME are present. In addition, as mentioned before, MF is not an essential parameter for method validation, therefore, measuring MF at different analyte concentrations may be unnecessary. Thus, MF is normally measured at an analyte concentration around mid-quality control (QC) sample. When using PCI profile convolution, the infusion rate is adjusted so that the signal intensity at the analyte-retention region is about the peak high of the mid-QC chromatogram. What about measuring MF for multiple lots of control matrix? The Crystal City conference report suggests that without isotopic-labeled IS, one should measure MF for six lots of matrix to make sure that ME are not variable across the lots. This suggestion seems to reflect that as long as ‘relative ME’ (i.e., variation of ME across different matrix lots) is not present, one needs not be too concerned [7]. One could argue that without isotopic IS, it is quite risky to accept a method with known ME greater than 15% of suppression or enhancement for a compound in development as ME is a finicky phenomenon that is likely to see variation of ME in incurred samples, especially in clinical studies.

Options to address ME

When one determines that there is an ionization ME in a method, what should be done? There have been discussions above about using separation techniques to reduce/remove ME. However, the following discussions will focus on a couple other practical areas related to ME.

■ Internal standards

It has been said that 'relative matrix effect' is more important, while 'absolute matrix effect' is of some concern (i.e., normal sense of ME). It should be pointed out that this approach is not prudent, at least for human samples that have wider degrees of matrix component variation, when stable-isotope IS is not used. Fortunately for regulated bioanalysis, most methods use stable-isotope IS, which correct for ME and many other perturbations during analysis. Sometimes, a stable-isotope IS does not compensate perfectly for the ME. In our experience, in those cases, the IS used were labeled with many (>4) deuteriums that eluted slightly earlier than the analyte, therefore experiencing different ME from the analyte. It is advisable to limit the number of deuteriums on the IS to four otherwise one may need to ramp the gradient in such a way as to merge the retention of the analyte and the IS, in order to prevent incomplete compensation of ME. In practice, in most cases it is not necessary to have more than M+4 label for small molecules, even when the molecule contains atoms such as chlorines or bromines, as long as one can place the label on the right fragment ion or one chooses a SRM transition (not necessarily an intense transition) away from potential crosstalk transitions. Given the fact that the IS compensation is critical for solving ME, it is very useful to determine IS-normalized MF, which is $MF(\text{analyte})/MF(\text{IS})$. An IS-normalized MF around 1.00 ensures good compensation by the IS regardless of what the MF of the analyte is. One may ask, even when the IS compensate for ME, at what MF value should the method be modified to reduce ME. To answer this question two factors need to be considered. One is the distortion of measurement precision and accuracy, and the other is quantitation limit reduction by ME. In our experience, we have seen methods easily exceed the precision accuracy requirement when several folds of suppression were exhibited, using stable-isotope IS. As for the quantitation limits, most of the modern methods have S/N ratios of the LLOQ samples much greater than 20:1, therefore, suppression of a couple folds rarely causes concern for reduction of LLOQ. However, in the practical sense, one should try to reduce ME to less than twofold of suppression or enhancements, even for stable-label IS methods. What about methods without IS or with analogue IS? It is the author's opinion that

it is better not to use an analogue IS especially when the IS elutes at a different retention time. Without the analogue IS, one needs to contend with ME in one retention region only. Adding an analogue IS eluting at a different retention time, not only does nothing to compensate the ME but adds its own ME potential. The value of an analogue IS for compensating volume variation during sample processing or HPLC analysis, and to a lesser degree for correcting MS sensitivity drop (due to dirty source and so on), has little worth considering the potential additional ME on the IS.

■ Different ionization techniques

There have been reports that ionization techniques other than ESI, such as APCI, APPI and EI, are less susceptible to ME [7,17–19]. APCI has been widely believed to be an effective alternative to ESI in solving ME. Data presented so far have not demonstrated that APCI is significantly better in the practical sense, than ESI. Remane *et al.* described a comprehensive list of compounds influenced by ionization suppression or enhancement due to co-eluting analytes [17]. The data showed marginally better ME behaviors for APCI versus ESI. Furthermore, some reports seem to indicate that APCI is prone to ionization enhancement [20–22]. The small advantage for APCI is further diminished by the fact ESI is much more sensitive for an average pharmaceutical compound, by approximately fivefold or more (based on some demo data obtained during instrument purchase). This is to say that if one dilutes with solvent, a final biological sample extract by fivefold (or whatever the factor of sensitivity advantage) in ESI analysis, and therefore the ME by ESI, will be reduced to a similar or even lower level than by APCI. Our experience in solving ionization suppression by iv. excipients bore this out. When we experienced severe suppression (several folds), we have not been able to use APCI to solve ME by simply switching from ESI to APCI.

Other means related to ionization techniques have similar pros and cons, as discussed in the comparison of ESI and APCI. Cappiello *et al.* has shown that EI is not sensitive to ionization ME [18]. However, the limitation of its sensitivity and usability to high-molecular-weight compounds prevents the EI technique from being used as a generic method in bioanalysis. APPI has been mentioned as less prone to ME than ESI [19]. The fact that dopants added to enhance ionization can alter an analyte response by 60-fold suggests

that APPI is not free of ME. In addition, it may also suffer similar limitations as EI to be used as a general technique. Reducing flow rate to the ion source has been reported to be effective in lowering ME in ESI [23,24]. Gangl *et al.* indicated that operating the LC flow at sub- μ l per min substantially reduces suppression, but the usefulness of nanoflow as a routine bioanalytical technique is limited in terms of throughput and ruggedness [23]. Kloepper *et al.* showed that reduction of HPLC flow to a practical rate of 50 μ l/min produced some improvement in reducing ionization suppression; however, the improvement was only incremental [24]. Overall, there has not been a magic bullet in using ionization-related techniques in removing ME in LC–MS bioanalysis.

Conclusion & future perspective

As discussed, much progress has been made in the understanding of ionization ME and their implications on quantitative bioanalysis. The bioanalytical field in the pharmaceutical industry is still moving towards a balanced understanding in terms of the extent and solution of ME. Because most pharmaceutical compounds tend to elute in the intermediate retention region on a commonly used C_{18} column, ME that are most severe in the very polar and nonpolar regions cause problems

less frequently in practice. Only when one analyzes iv. samples containing surfactant excipient does the need to address ME become routine. Fortunately, even for those cases, modern triple-quadrupole MS have become so sensitive, allowing these samples to be diluted by more than 100-fold (e.g., at least tenfold by control plasma and tenfold by solvent in protein precipitation), which make ME a nonissue in these cases. In a sense, the push for increased MS sensitivity is no longer for the sake of analyte concentrations (with a few exceptions), but rather for the flexibility of diluting samples to avoid ME. Moreover, in another sense, ME are one of the difficult issues that require in-depth expertise to solve, preventing bioanalysis from becoming a totally commoditized service that would put us bioanalytical scientists out of work any time soon.

Financial & competing interests disclosure

The author has no relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript. This includes employment, consultancies, honoraria, stock ownership or options, expert testimony, grants or patents received or pending, or royalties.

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Executive summary

Background

- Ionization matrix effects (ME) need to be addressed in quantitative bioanalysis, especially in regulated bioanalysis.

Source of ionization ME

- Sources of ME are much more than phospholipids.

Methods of ME assessment

- Quantifying ME by comparing response of analyte spiked in blank matrix extract against neat solution is sometimes distorted by nonspecific binding. Measurement using infusion profile convolution removes such a problem.

Options to address ME

- Solving ME problems by sample cleanup, HPLC separation or different ionization techniques can be effective to some extent. The use of a good stable isotope-labeled internal standard (IS) is a reliable and simple way to mitigate ME.

Internal standards

- Use of nonisotopic-labeled IS generally does not help to reduce/eliminate ME, but rather increases the chances of ME if the IS elutes at a different retention time.

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